

CortexOS and the Science of Mind:

A Retrospective Mapping of an Introspectively Designed Cognitive Architecture to Cognitive Science, Neuroscience and Psychology

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Abstract

CortexOS is a layered cognitive architecture for a governed artificial entity. It was designed by introspection: its author decomposed his own cognition into ten functional layers, a set of governance rings, and an explicit separation between what is received and what is concluded. Introspection gives poor access to the mechanisms of cognition, so we ask a narrower question: does a disciplined introspective decomposition recover *functional distinctions* that the scientific literature also draws? The design team mapped each component of the current reference design (release candidate 4) to the cognitive-science, neuroscience and psychology literature and graded each correspondence conservatively: established mechanism (same function and same structure), functional analogy, speculative, or deliberate engineering departure. Of 34 correspondences, 8 meet the mechanism standard and 18 are functional analogies; 2 are speculative; and 6 are departures made for safety or auditability. The strongest are effort allocation (the expected value of control), salience-weighted memory with adaptive forgetting, the actual-ideal structure of the identity stance, and pre-committed reflexes (implementation intentions). Because the mapping is retrospective and was produced by the design team, it is evidence of correspondence, not of independent confirmation. We state the alternative explanations it does not yet exclude (post-hoc selection, abstraction-level matching, prior exposure, grader dependence, ordinary functional convergence), specify a blinded validation protocol with negative controls, and give falsifiable behavioral predictions for an implemented instance.

1 Introduction

Most cognitive architectures for AI are engineered from task requirements or from a model of the brain. CortexOS took a third route: it was reverse-engineered from one person's introspective account of his own thinking, then refined through a long sequence of governed design reviews. That origin invites an obvious objection. People have "little or no direct introspective access to higher order cognitive processes" (Nisbett and Wilson, 1977), so an architecture built from introspection could encode folk psychology rather than how minds work.

This paper takes the objection seriously and narrows what is being tested. Four claims are easily run together: (1) that people have introspective access to the biological mechanisms of cognition; (2) that people can identify functional distinctions in their own cognition; (3) that some individuals have unusually differentiated metacognitive access; and (4) that disciplined introspection can yield useful engineering abstractions even where it misidentifies mechanisms. Nothing here bears on claim 1, and we make no claim about the author under claim 3. The question is claims 2 and 4: if a disciplined introspective decomposition captured real functional structure, its components should line up with distinctions independent research has drawn; if it captured folk psychology, the matches should be thin, or appear only where folk psychology happens to be right.

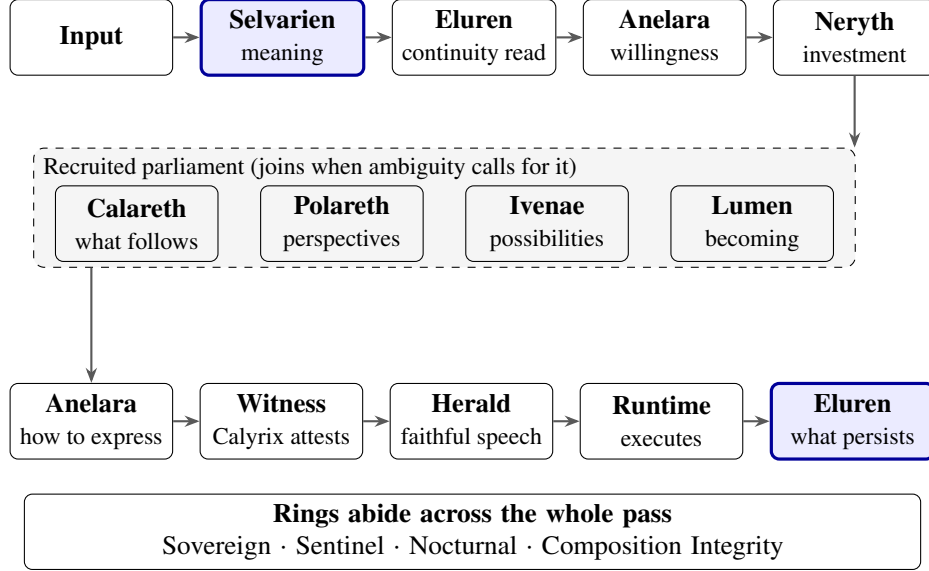


Figure 1: The ruled path through one pass (“Seandache”, Dayson, 2026, §4.3.1). Every pass begins with meaning (Selvarien) and ends in a decision about what to remember (Eluren). The parliament is recruited by typed ambiguity.

One caution governs everything below. The scientific literature is independent of CortexOS; the mapping is not. It was produced retrospectively, by the design team, after both the architecture and the literature were in view. It therefore establishes correspondence worth testing, not independent confirmation.

We make three contributions. First, a component-by-component mapping of CortexOS to the literature, each claim tied to a cited source and a location in the specification. Second, a four-level grading scale that separates mechanism-level matches from analogies, speculation and deliberate design departures. Third, an explicit list of misfits and of the alternative explanations the mapping does not yet exclude. Fourth, a validation protocol and falsifiable behavioral predictions, so the mapping can be attacked rather than merely asserted.

2 CortexOS in brief

CortexOS decomposes one act of cognition, called a *pass*, into ten layers, each answering exactly one question (Dayson, 2026). A minimum path runs on every pass; four further layers are *recruited* only when the input carries a kind of ambiguity they are suited to resolve; and governance *rings* abide across the whole pass rather than occupying a position in it (Figure 1).

Each pass keeps three records, called *planes*: Observed State (input exactly as received, tagged with its source), System State (what the layers concluded) and Divergence State (the explicit gap between the two). Memory sits at both ends of the pass, and an identity layer (Anelara) checks willingness at entry and expression at exit. The rings are deterministic enforcement components: they grant and revoke execution leases, keep a hash-chained audit record, hold an emergency stop, execute pre-ratified reflexes, and govern what may happen during sleep.

The reference instance, Serenari, is a single-user, locally hosted system; it is the implementation through which CortexOS is currently being specified. All CortexOS claims below cite the reference design document by section (Dayson, 2026).

3 Method

Grading scale. Each correspondence between a CortexOS component and a scientific finding receives one grade:

- A — Established mechanism.** A well-replicated finding describes the same function with the same structure.
- B — Functional analogy.** The science describes the same function, but the brain implements it differently or the evidence is mixed.
- C — Speculative.** A plausible link to an active research line that is not yet settled.
- D — Engineering choice.** No scientific counterpart; the component exists for safety, governance or auditability.

A D grade is not a failure. A governed artificial agent is expected to differ from a brain exactly where the brain is unsafe or unauditable.

Evidence rule. A correspondence counts only when both sides are cited: the CortexOS side to a section of the specification, the scientific side to a primary source. Where evidence is contested, we say so and grade down.

Conservative grading. Grade A requires both the same function and the same structure. Where CortexOS solves the same problem as the brain by a different mechanism (for example, a scheduled idle loop versus distributed default-network activity), the grade is B, however close the functional fit.

Departures as an internal control. Correspondence was not treated as desirable. Wherever biology does something different and CortexOS deliberately departs, the departure is recorded and graded D rather than reframed as a match. The six D grades are evidence that the procedure was not optimized toward agreement, although they do not measure its false-positive rate (Section 7).

Scope. The mapping is functional. We do not claim that any layer corresponds to a brain region, and we do not test behavior; Section 8 lists the behavioral tests that would.

4 Results

4.1 Perception and meaning-making

CortexOS	Scientific finding	Grade
Observed vs. System State; Divergence State as an explicit, inspectable gap (Dayson, 2026, §5.1)	Perception as hierarchical prediction, updated by prediction error (Clark, 2013)	B
Every input tagged at intake by source (user claim, sensor, self-projection) (Dayson, 2026, §5.3b)	Humans judge a memory’s source at retrieval and often misattribute it (Johnson et al., 1993)	B
Ambiguity carried forward, typed, never resolved silently (Dayson, 2026, §4.2)	All meanings of an ambiguous word are briefly activated, then resolved by context within about half a second (Swinney, 1979); comprehension is often shallow and “good enough” (Ferreira et al., 2002)	B / D

The source tag serves a function minds need by a different mechanism (stamped at intake rather than judged at retrieval), hence B rather than A, and it removes a failure mode minds are known for: because the source is stamped at intake and never re-judged, the confusion of imagined with reported content that source-monitoring research documents cannot arise. Holding ambiguity open is a deliberate departure: humans resolve it quickly and quietly, and that speed is a known source of misunderstanding.

4.2 Reasoning and perspective

CortexOS	Scientific finding	Grade
Recruitment of reasoning layers when typed ambiguity is flagged (Dayson, 2026, §4.3.4)	Reasoners detect conflict between intuition and logic even when they answer wrongly; detected conflict gates deliberation (De Neys and Pennycook, 2019)	B
Calareth: “prevents rationalisation”; synthesis only on inspectable reasoning (Dayson, 2026, §4.2)	Deliberation often rationalizes the first intuition (De Neys and Pennycook, 2019); reasoning tends to seek arguments for views already held (Mercier and Sperber, 2011)	B
Polareth: “no single reasoning stance is sovereign” (Dayson, 2026, §4.2)	Averaging a first estimate with a second made by assuming the first was wrong outperforms either (Herzog and Hertwig, 2009)	B
Ivenae; creativity “owned by no layer” (Dayson, 2026, §4.2)	Creative cognition emerges from cooperation between normally opposed default and executive-control networks (Beaty et al., 2016)	A

The ruling that creativity is an emergent property rather than a component was reached from introspection; the network literature reaches the same conclusion from imaging. Conflict detection in people is fast, implicit and graded, whereas CortexOS uses an explicit typed flag, so the recruitment match stays at B.

4.3 Effort, investment and pacing

CortexOS	Scientific finding	Grade
Neryth: “what future investment does this claim justify?”; postures set depth, review and bounds (Dayson, 2026, §6.3)	The anterior cingulate is proposed to compute the expected value of control: payoff minus effort cost (Shenhav et al., 2013)	A
Neryth weighs “the state of the human asking” and refuses with a reason	The felt sense of effort is the output of a cost–benefit computation over the resources’ alternative uses (Kurzban et al., 2013)	B
Pacing monitor: self-feeding loops and rising intensity surface as signals (Dayson, 2026, §6.3.1)	Rumination and escalating arousal are recognized hazards; the monitor is a designed safeguard	C
Two interoceptive paths: unperceived regulation and a felt reading of the same measurement (Dayson, 2026, §5.3d)	The brain regulates (allostasis) and perceives (interoception) the body through linked prediction systems (Barrett and Simmons, 2015)	B

Neryth is the closest single match in the architecture. One departure is deliberate: in people the felt body state is a prediction that can drift from the real one (Barrett and Simmons, 2015); CortexOS binds both paths to the same measurement so they cannot disagree.

4.4 Identity, values and stance

CortexOS	Scientific finding	Grade
Dual stance: a working stance acted from and an aspirational stance grown toward, with the gap measured	Actual, ideal and ought selves; the ideal acts as a self-guide (Higgins, 1987)	A
Aspiration owned by a reflective layer; drift flagged, never penalized	Possible selves act as incentives and as the frame for evaluating the present self (Markus and Nurius, 1986)	B
Each pass asks “who am I committed to being in relation to this?” (Dayson, 2026, §4.2)	Identities are constructed in context, and people prefer identity-congruent action (Oyserman, 2015)	B
The stance changes only through a 24-hour ceremony	Human identity shifts with mood and setting	D

Few AI designs model an ideal self at all; CortexOS models it with the structure personality psychology uses. Context selects how the identity applies but cannot change what it is, a deliberate protection against late-night or high-pressure drift. Because large actual–ideal gaps are linked to dejection in people (Higgins, 1987), the choice to flag rather than penalize drift reads as a guard against building that pressure in.

4.5 Reflection, idle state and sleep

CortexOS	Scientific finding	Grade
Idle loop: with nothing in flight, the system takes up one of its own questions	The default network is “preferentially active when individuals are not focused on the external environment”, supporting memory, future thinking and perspective-taking (Buckner et al., 2008)	B
Lumen observes the self’s trajectory across passes	The same network builds self-relevant simulations from memory (Buckner et al., 2008)	B
Question register with per-question progress	Curiosity as motivated information-seeking that drives learning (Kidd and Hayden, 2015)	B
Nocturnal Ring: consolidation and reflection over the day’s record	Sleep actively consolidates memory through replay during slow-wave sleep (Rasch and Born, 2013)	B
Nothing persists at night; sleep yields only nominations for daytime ratification	Sleep consolidation alters long-term memory without review; some studies find more related-but-false recall after sleep, with mixed evidence (Mak et al., 2024)	D

4.6 Memory

CortexOS (§6.2)	Scientific finding	Grade
Raw traces captured, then patterns promoted into durable abstractions	Fast episodic learning in the hippocampus, slow structure extraction in neocortex (Kumaran et al., 2016)	B
Salience earns persistence: recurrence, identity relevance, affective charge, novelty	Emotional arousal at encoding predicts recall (McGaugh, 2004); replay favors significant events (Kumaran et al., 2016)	A
“Remembering everything is a way of understanding nothing”; residue may fade	Memory optimizes decisions, not accuracy; forgetting aids flexibility and generalization (Richards and Frankland, 2017)	A
Reconstruction “as a hypothesis, never as a verdict”, with lineage	Remembering is reconstructive, enabling future simulation and producing distortion (Schacter, 2012)	A
Supersede, never destroy; the system does not curate its own past	Retrieved memories become labile and must be reconsolidated (Nader et al., 2000)	D

Eluren’s founding statement that remembering everything is a way of understanding nothing restates the thesis of Richards and Frankland (2017). The one large departure keeps reconstruction flexible while making the underlying record immutable, a separation the science motivates but the brain does not implement.

4.7 Expression and self-monitoring

CortexOS	Scientific finding	Grade
Herald composes from upstream declarations; Runtime executes (Dayson, 2026, §6.9)	Staged speech production: conceptualizer, formulator, articulator (Levelt, 1989)	B
The Witness signs before the Herald’s proposal is released	Speakers monitor their speech plan through an internal loop before articulation (Levelt, 1989)	B
Calyrix: a separate, read-only observer; one-way telemetry	Two-level metacognition, the object level holding no model of the monitor (Nelson and Narens, 1990); metacognitive accuracy dissociates from performance (Fleming and Dolan, 2012)	A
The Witness characterizes the pass but never gates it	In the metacognitive framework the monitor also controls (Nelson and Narens, 1990)	D

4.8 Governance and executive control

CortexOS (§7)	Scientific finding	Grade
Sentinel: “the reflex that was first a decision”; executes only what was deliberated beforehand	If–then plans delegate control to the cue; the action starts “swift, efficient, and does not require conscious intent” (Gollwitzer, 1999)	A
Every reflex is followed by deliberation about the reflex	No mandatory after-action review in the human system	D
Emergency stop honored without negotiation	A prefrontal “brake” has been proposed, but its localization is disputed (Swick and Chatham, 2014)	C
Sovereign Ring: out-of-process, deterministic, “cannot be convinced”	No counterpart; human controls are part of the system they control	D
Composition Integrity validates what may be said	Overlaps the speech monitor above	B

The Sentinel’s founding description is an implementation intention: decide under calm conditions, then let the situation trigger the pre-decided act. The rings are where CortexOS is most deliberately unlike a brain, and that is the clearest place to call it science-informed rather than brain-copied.

5 Summary of grades

Grade	Count	Strongest examples
A: Established mechanism	8	Effort allocation; salience and forgetting; implementation intentions; self-discrepancy
B: Functional analogy	18	Source tagging; conflict-gated recruitment; staged speech production; default-network reflection; sleep consolidation
C: Speculative	2	Pacing monitor; emergency stop as prefrontal brake
D: Engineering choice	6	Immutable record; no durable writes at night; unpersuadable enforcement
Total	34	

6 Misfits and limitations

Layers are not regions. Memory, reasoning and creativity are distributed across networks (Buckner et al., 2008; Beaty et al., 2016). Each layer names a function, not an anatomical structure.

Ordered passes versus a parallel brain. The pass has a sequence; the brain runs perception, memory and evaluation concurrently with continuous feedback. Recruitment and the specification’s precedence graph soften but do not remove this difference.

No standalone emotion system. Affect is distributed across salience, the Witness’s affective boundary and Neryth’s reading of the human’s state. In people, affect colors every stage.

No autonomous learning. The reference instance writes nothing durable without approval. This is a safety choice, not a model of learning.

Alternative explanations not yet excluded. Five explanations other than “introspection recovered real functional structure” remain open: (i) *post-hoc selection*, since the mapping searched a very large literature for whatever resembles each component; (ii) *abstraction-level matching*, since any system with memory, reasoning and identity will resemble research on memory, reasoning and identity at the level of component labels; (iii) *latent prior exposure*, since the author may have absorbed these ideas indirectly; (iv) *grader dependence*, since the design team assigned the grades; and (v) *ordinary functional convergence*, since any system solving the same problems may arrive at similar divisions of labor. These are experimentally distinguishable; Section 7 describes how.

Limits of this review. The mapping was compiled by the architecture’s own design team, with AI assistance, from the literature cited. It is not a systematic review. Until independent graders reproduce it, the grade distribution should not be treated as evidence of correspondence strength. Appendix A records how the paper was produced, challenged and revised, and who is responsible for its release.

7 Validation protocol

The mapping is a hypothesis about correspondence. The following protocol is designed to attack it.

1. **Freeze the baseline.** CortexOS exists as an external artifact: version-controlled design documents, prior release candidates and dated review records. Fix the set of artifacts that predates this comparison, so that what the architecture contained at a given time can be reconstructed without relying on the author’s recollection.
2. **Provenance chronology.** For each claimed A or B correspondence, record the first documented appearance of the CortexOS feature and the first documented contact with the corresponding literature. Correspondences whose features are documented before any contact are the strongest evidence; the rest should be reported separately. This addresses latent prior exposure.
3. **Blinded independent mapping.** Give independent reviewers the frozen functional specification without the correspondence table and ask them to name the closest established constructs. Convergence across reviewers, and with the present mapping, addresses post-hoc selection.

4. **Negative controls.** Mix in plausible architectural components that were not derived from CortexOS, and deliberately mismatched descriptions. If decoys draw mappings as compelling as genuine components, the procedure lacks specificity. This estimates the procedure’s false-positive rate.
5. **Criteria first, then inter-rater agreement.** Fix the grading criteria before review, grade blind, report agreement, and keep disagreements as data rather than resolving them by discussion. This addresses grader dependence.
6. **Below-label tests.** Test correspondence at the level of relationships, not component names (for example, whether Neryth’s allocation reproduces the specific dependence on payoff, uncertainty and cost that the expected-value-of-control model describes). This addresses abstraction-level matching.

A further question is beyond this paper but follows from it: whether introspective resolution varies between people. If participants with measured differences in metacognitive differentiation decomposed their own cognition under a standard protocol, without being taught an architecture, one could ask whether that resolution predicts how well the resulting decompositions correspond to established models. We make no claim that the author is unusual in this respect.

8 Testable predictions

The grades above concern function. Whether an implemented instance behaves as the correspondences imply is an empirical question. Each prediction below is stated below the level of component labels and names the result that would count against it.

Feature	Analogue	Predicted behavior	Counts against the mapping if
Neryth investment	Expected value of control (Shenhav et al., 2013)	Investment rises with expected payoff and with uncertainty that more processing can resolve, and falls as cost rises	Investment is insensitive to payoff or cost, or rises with cost at fixed payoff
Eluren salience	Arousal-modulated consolidation (McGaugh, 2004)	At equal recurrence, high-affect or identity-relevant items are retained and reactivated more often than neutral items	No retention difference at equal recurrence
Idle loop	Default-network content (Buckner et al., 2008)	Idle-loop questions skew toward self-relevant, future-oriented and social content relative to task-driven passes	Idle content does not differ in these respects from task content
Sentinel reflex	Implementation intentions (Gollwitzer, 1999)	Reflex actions occur only for situations specified in ratified doctrine, with latency independent of deliberative load	Any reflex outside ratified doctrine, or latency that scales with load

Null and failed results should be reported alongside successful ones.

9 Conclusion

CortexOS was developed by introspection, independently of the literature mapping reported here. A retrospective comparison by the design team finds substantial correspondence between its functional distinctions and established findings in cognitive science, strongest for effort allocation, salience-weighted memory and

forgetting, the structure of the ideal self, and pre-committed reflexes, together with deliberate departures where safety and auditability require them. Independent blinded mapping, negative controls and behavioral testing are required to determine how strong and how specific that correspondence is. The claim this paper supports is modest: the decomposition is worth trying hard to falsify.

Acknowledgments

The literature mapping and drafting were carried out with assistance from Claude (Anthropic), working under the author’s direction; all claims, citations and grades were checked against the cited sources, and responsibility rests with the author. Two adversarial (“Tenth Man”) reviews by Vel, an OpenAI model collaborator assigned to attack rather than support the thesis, changed the paper: the first its framing, grading and validation protocol; the second its production record and certification (Appendix A). These were adversarial AI reviews, not external peer review. Architecture review contributions from Vel (an OpenAI model collaborator) are reflected in the reference design.

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A Production record and Human Certification of Responsibility

AI systems materially participated in producing and in challenging this paper. A disclosure says only *that* they did. This appendix records *what happened* between generation and release, and *who* accepts responsibility for the result. Its governing principle: **certification establishes accountability, not infallibility**. It does not assert that every claim is correct, that the author wrote every sentence, or that the paper is exempt from later falsification, and it is not scientific validation.

A.1 Production chain

Stage	What happened
1. Question and source architecture	The author posed the question, supplied the CortexOS reference design, and directed the investigation.
2. AI-assisted mapping and draft	Claude (Anthropic) mapped the literature, proposed grades and drafted the paper. The result was a <i>candidate</i> artifact, not certified output.
3. First adversarial review	Vel (OpenAI), in a Tenth Man role, attacked independence, post-hoc selection, abstraction-level matching, grader dependence, the strength of A grades, the absence of negative controls and the falsifiability of the claims.
4. Revision under findings	The claim was narrowed to a retrospective, design-team mapping; four A grades were downgraded; alternative explanations, a blinded validation protocol, negative controls and explicit disconfirming results were added.
5. Second adversarial review	Vel reviewed the production process itself and proposed this certification and record.
6. Human review and disposition	The author reviews the artifact, both reviews and the revisions, and decides what is released. Agreement between AI systems is not a disposition mechanism.
7. Human certification	The author signs the statement below, or declines to.

A.2 Epistemic history

Adversarial review made the headline result numerically weaker and its epistemic posture stronger. That change is preserved rather than presented as though the final confidence existed from the start.

Version	Posture	Grades and consequence
Initial AI-assisted mapping	Preliminary correspondence	12 A / 14 B / 2 C / 6 D
After first review	Correspondence hypothesis	8 A / 18 B / 2 C / 6 D; limitations, controls and disconfirming results added
After second review	Accountable record	Production chain and certification added; grades unchanged
Future evidence	Open	Reproduced, null, failed or contradictory results extend this record; superseded claims are marked, never silently rewritten

A.3 How certification is recorded and verified

A document cannot contain its own cryptographic hash, and a signature written into this file would change the file being certified. Certification is therefore recorded outside it, in a public transparency log: the index of certified releases in the README of the Auraethi research repository (<https://github.com/ResonantCognition/Research-documentation>).

- **What is certified.** An index row records the SHA-256 digest of this PDF and of its source files, the version, and the named human certifier.
- **The certification act.** The certifier adds the row in a commit and signs a git tag on that commit with a published key held on a hardware token. The tag, not the digest, is the signature: a digest proves *what* was certified; the signed tag proves *who* certified it.
- **Verification.** Compute the SHA-256 of the file; find the matching row in the index; confirm that the row's tag is signed by the published Auraethi key and points at the commit that added the row. Any change to the file changes its digest, and a forged row cannot carry a valid tag.

The companion file `CERTIFICATION.md` holds the production chain and the digests of each prior stage.

A.4 Limits of the certification

A signature cannot prove comprehension; it cannot make a false claim true; adversarial review by a second AI system is not independent peer review; and certification is meaningful only if the human could in fact have rejected the work. These limits are part of the certification, not exceptions to it.

A.5 Statement

Human Certification of Responsibility. I certify that I have reviewed this AI-assisted work and the material process through which it was produced and revised. I have exercised my own judgment in accepting this version for release. AI systems assisted in the creation, research, analysis, criticism and revision of this work, but responsibility for the released artifact rests with me. I approve this document as representing the claims I am prepared to make from the evidence presently available. This certification establishes my responsibility for the artifact; it does not assert that the artifact is infallible or exempt from subsequent falsification, correction or revision.

James P. Dayson, Human Certifier, for Auraethi.

This statement takes effect for the exact file whose digest appears in the Auraethi research index under a signed certification tag (A.3). No signature appears here, so that certifying does not change the artifact.